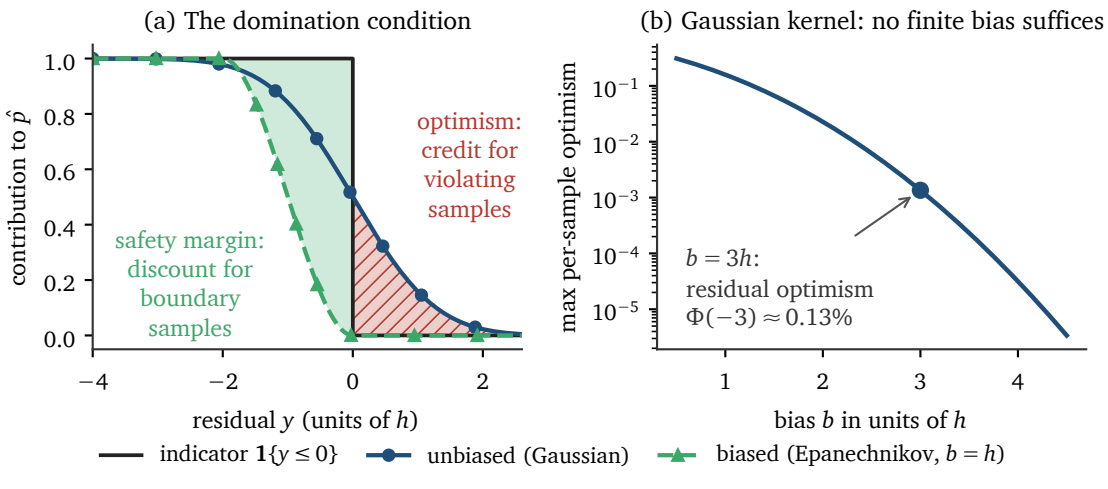
(a) The domination condition
contribution to $\hat{p}$
1.0
0.8
0.6
0.4
0.2
0.0
optimism:
credit for
violating
samples
safety margin:
discount for
boundary
samples
−4
−2
0
2
residual y (units of h)
(b) Gaussian kernel: no finite bias suffices
max per-sample optimism
10^{-1}
10^{-2}
10^{-3}
10^{-4}
10^{-5}
$b = 3h$:
residual optimism
$\Phi(-3) \approx 0.13\%$
1
2
3
4
bias b in units of h
indicator $\mathbf{1}\{y \le 0\}$
unbiased (Gaussian)
biased (Epanechnikov, $b = h$)